
Report: **AM2024-0074**

TEM investigation on particle samples of natural origin

Short Summary 5

Client:

Curia Germany GmbH
Industriepark Höchst D569
65926 Frankfurt am Main

Contractor:

Fraunhofer-Institut für Silicatforschung ISC
Zentrum für Angewandte Analytik ZAA
Neunerplatz 2
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Project Management:

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Datum: 2024.10.23 12:38:00 +02'00'

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Documentation Overview

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ISC Offer Number

2210-2024-51000-45-02801-00010 and
2210-2024-51000-45-02801-00019

Order of the Client

signature on offer 2210-2024-51000-45-02801-00010
dated 26 July 2024 and
signature on offer 2210-2024-51000-45-02801-00019
dated 09 September 2024

Place of Performance and Processing Time

Place of Performance

Center for Applied Analytics
Fraunhofer Institute for Silicate Research
Neunerplatz 2
97082 Würzburg
Germany

Processing Time

Sample receipt	29.07.2024 & 28.08.2024
Begin of the analysis	22.10.2024
End of the analysis	22.10.2024
Date of report	23.10.2024

Sample Overview

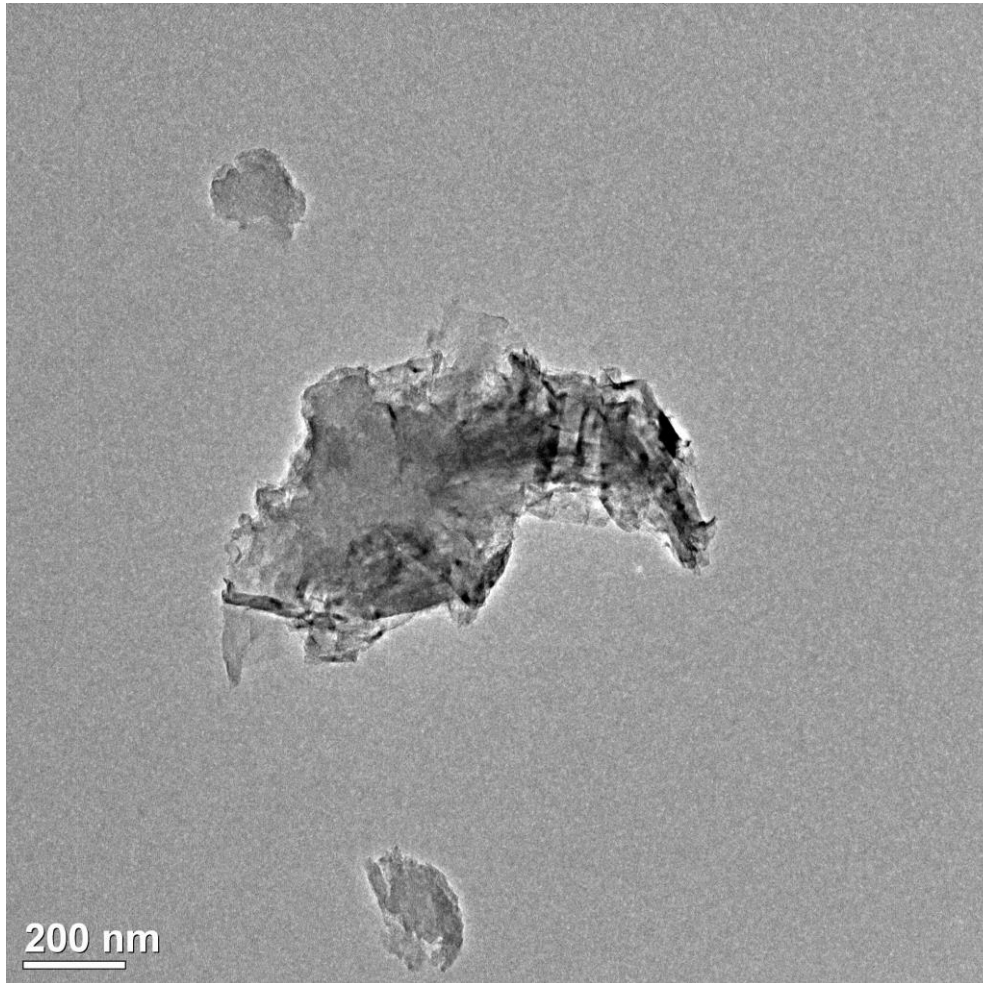
Sample No.	Description
P02	Graphene – Batch: TB1137-1 – Date: 24.07.2024
P03	Graphene – Batch: MB2976A – Date: 23.08.2024

Procedure of Testing

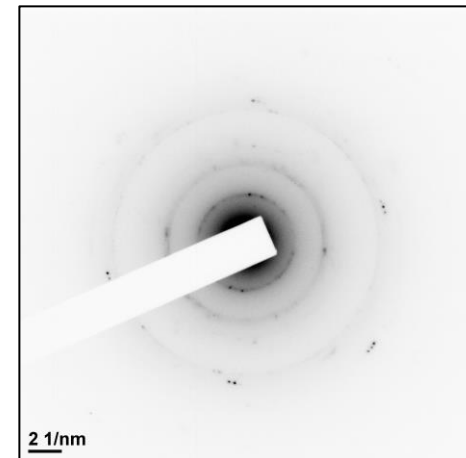
- The received samples were ground before a small amount of the powder was dispersed in 1 part IPA and 4 parts of water.
- The dispersions were treated with an ultrasonic bath for 15 min.
- The dispersions were dripped onto a TEM finder grid with a Formvar foil.
- The samples were dried on air without removing any excess liquid and afterwards transferred into the TEM.
- The investigations of the particles were done in TEM brightfield and darkfield mode. Diffraction analysis was done to check for crystallinity.

TEM investigation

Sample P02 (TB1137-1)

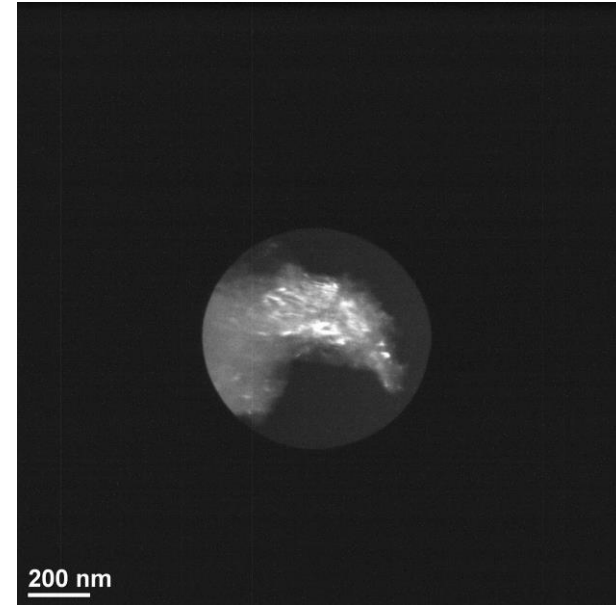
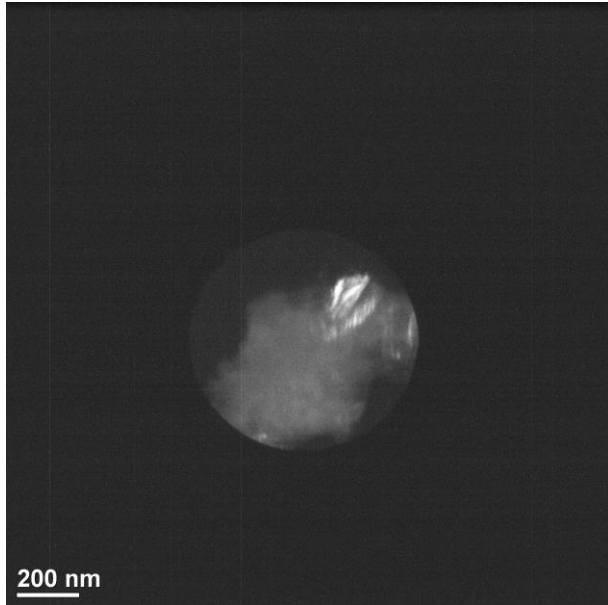


- This TEM brightfield image shows a particle of sample P02 at a magnification of 10.000 times.
- It exhibits a rather flat structure.
- The diffraction pattern shows clear reflexes indicating a crystalline structure.



TEM investigation

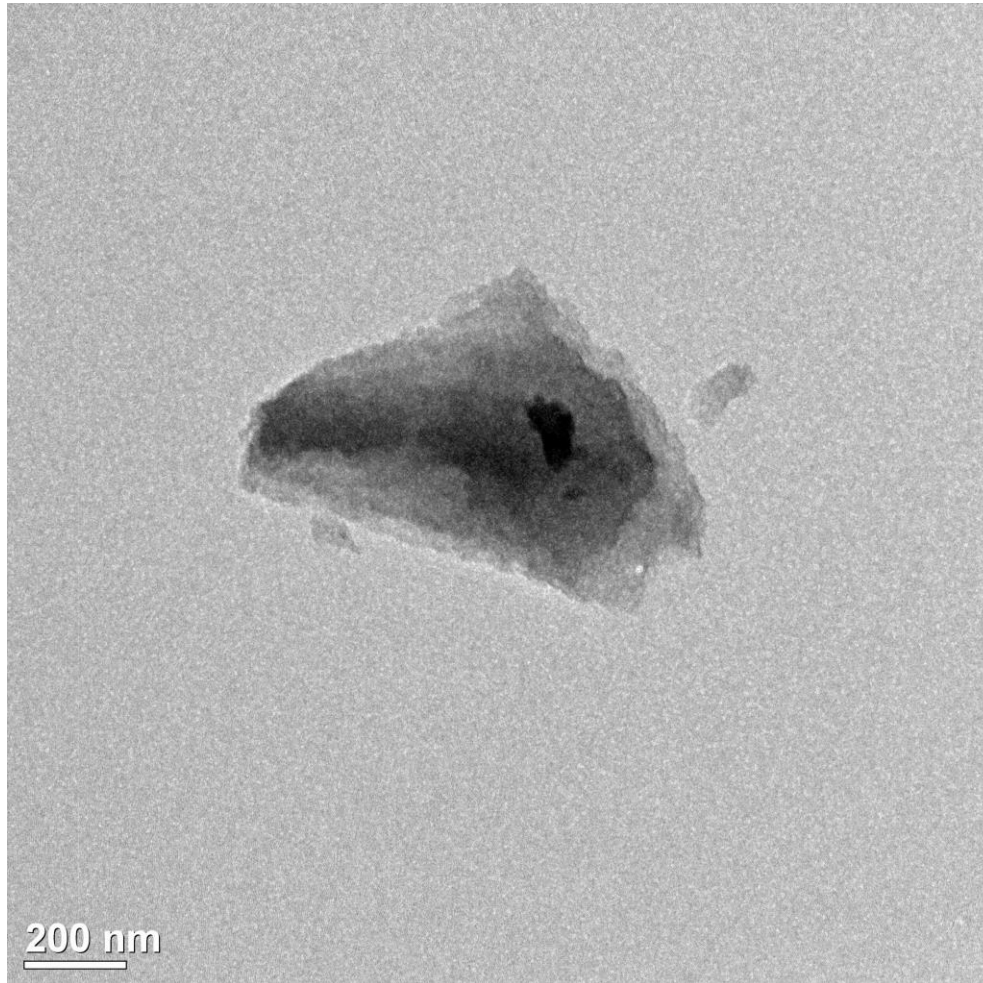
Sample P02 (TB1137-1)



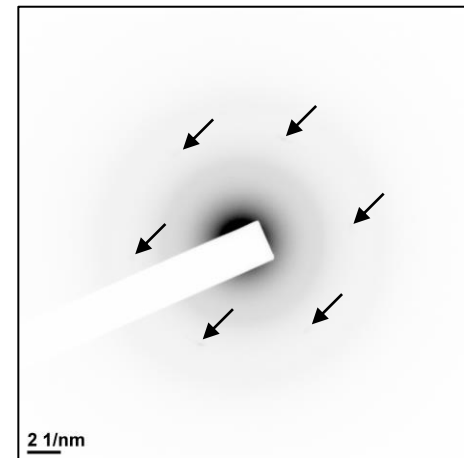
- These two darkfield images show crystalline regions of the previous particle.

TEM investigation

Sample P02 (TB1137-1)

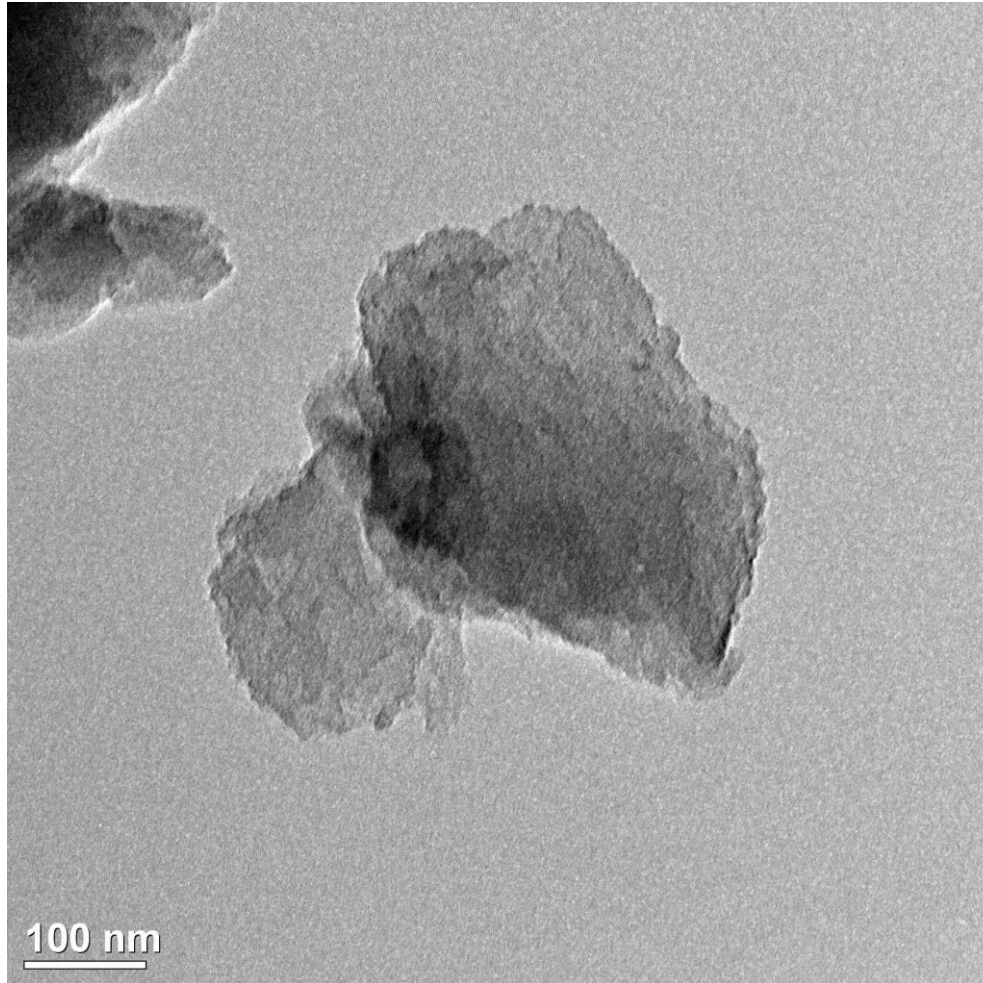


- This TEM brightfield image shows a particle of sample P02 at a magnification of 10.000 times.
- The diffraction pattern shows weak reflexes. Some of them are marked with arrows

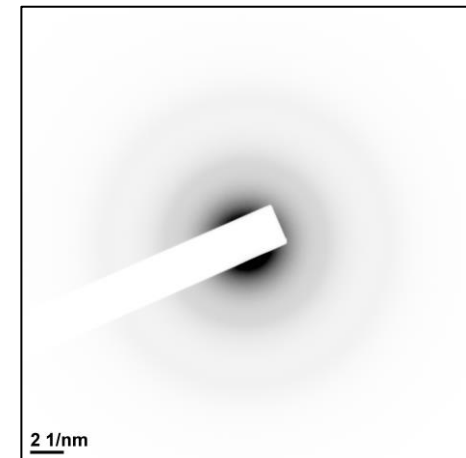


TEM investigation

Sample P02 (TB1137-1)

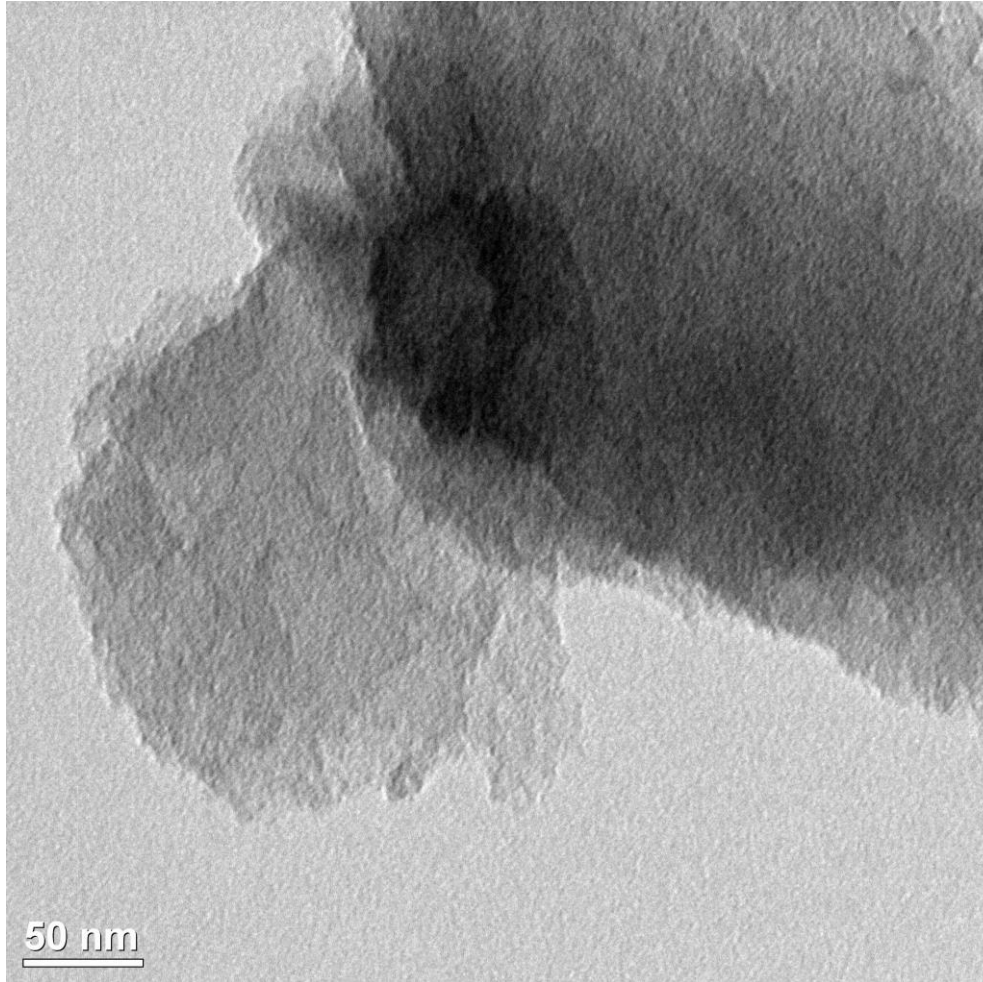


- This TEM brightfield image shows a particle of sample P02 at a magnification of 25.000 times.
- It also exhibits a rather flat structure.
- The diffraction pattern shows no pronounced reflexes.



TEM investigation

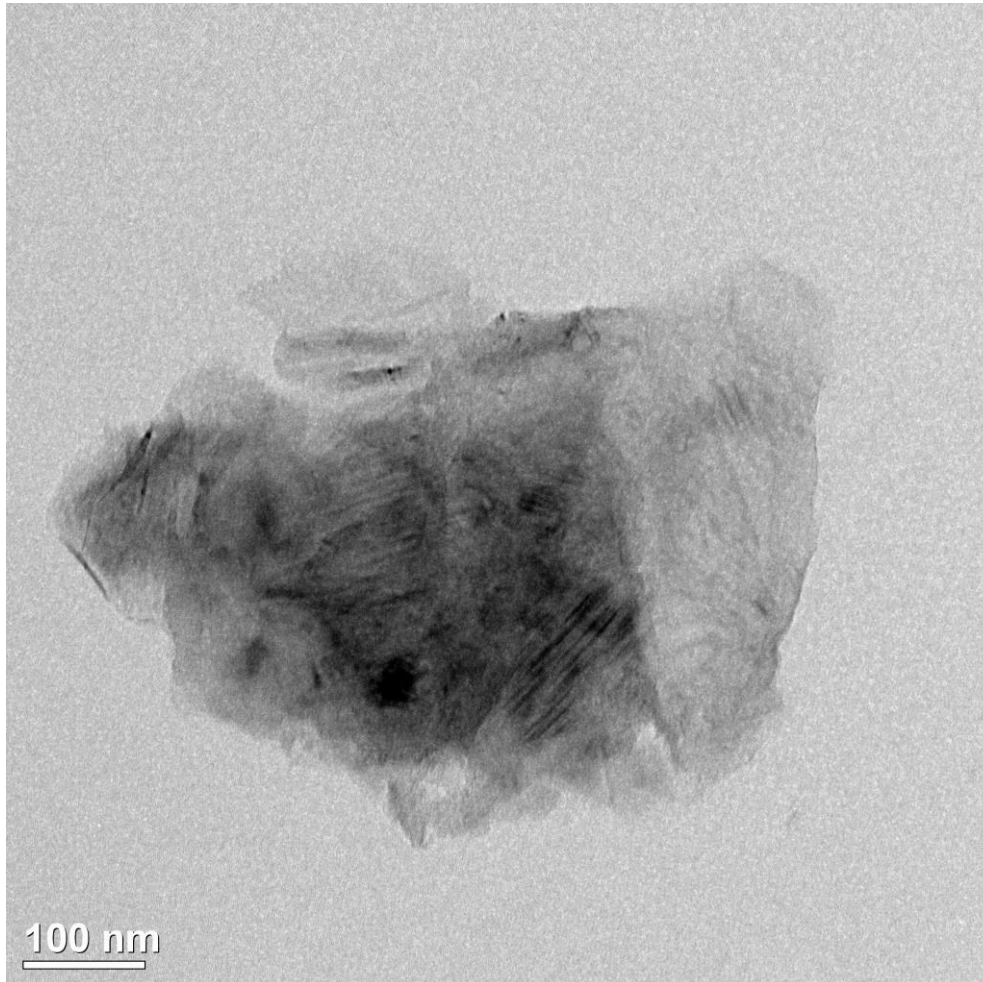
Sample P02 (TB1137-1)



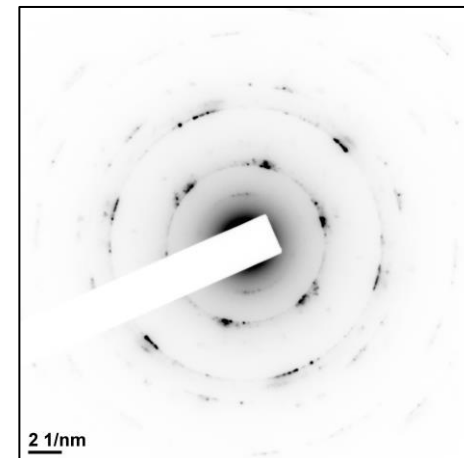
- This TEM brightfield image shows the previous particle at a magnification of 50.000 times.

TEM investigation

Sample P02 (TB1137-1)

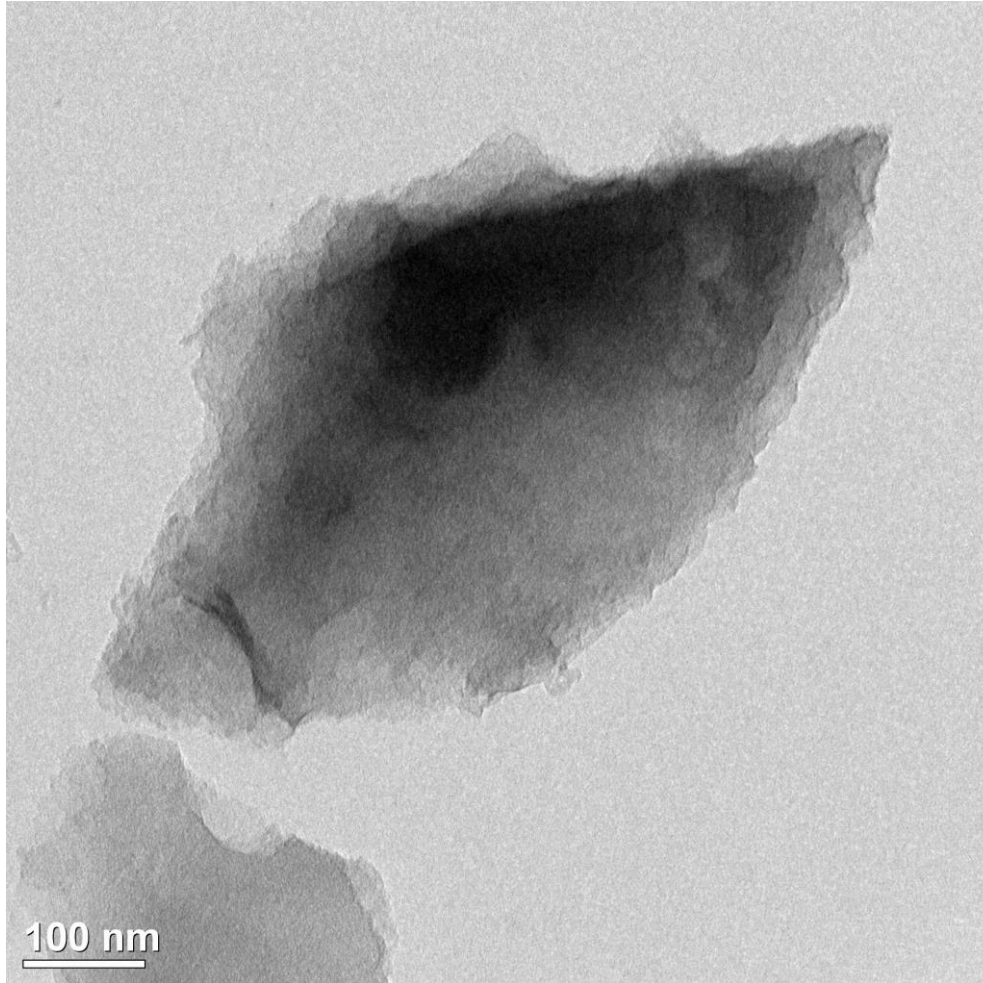


- This TEM brightfield image shows a particle of sample P02 at a magnification of 25.000 times.
- It also exhibits a rather flat structure. Hints of crystallinity are already visible.
- This is confirmed by the diffraction pattern that shows pronounced reflexes.

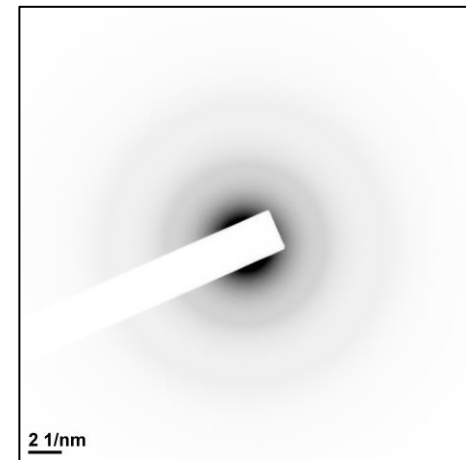


TEM investigation

Sample P03 (MB2976A)

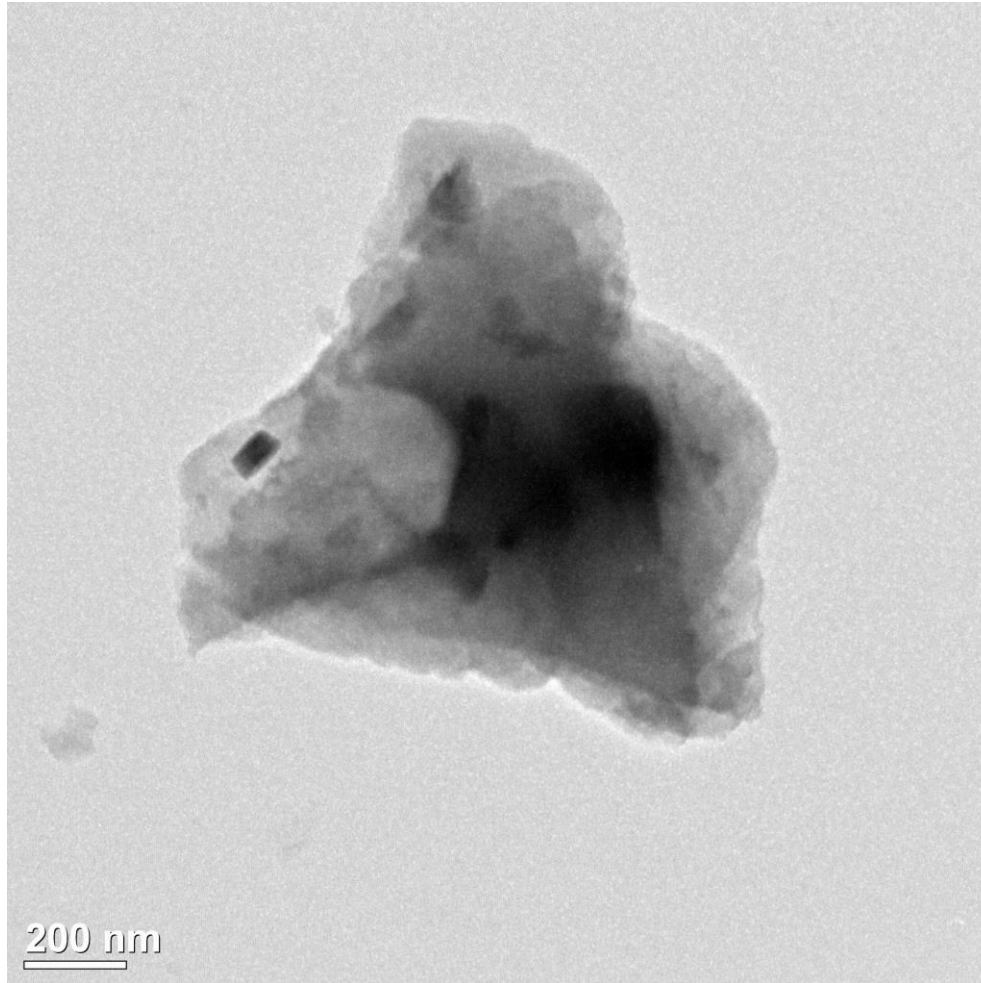


- This TEM brightfield image shows a particle of sample P03 at a magnification of 25.000 times.
- The diffraction pattern does not show crystalline reflexes.

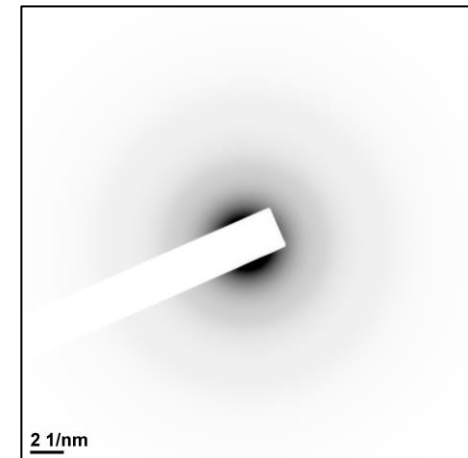


TEM investigation

Sample P03 (MB2976A)

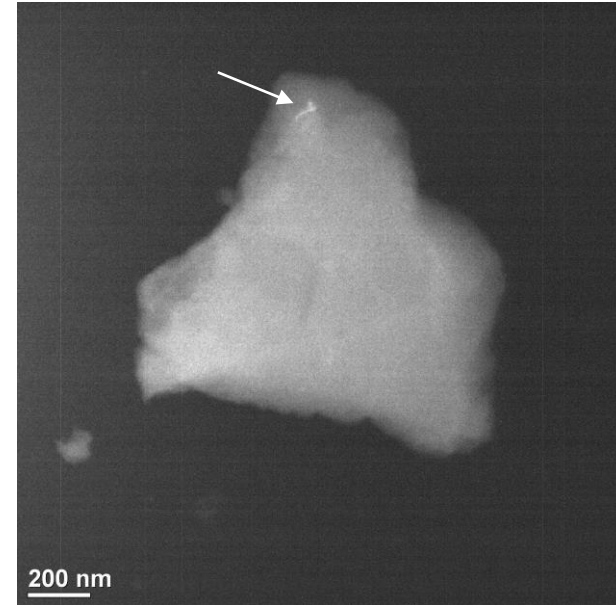
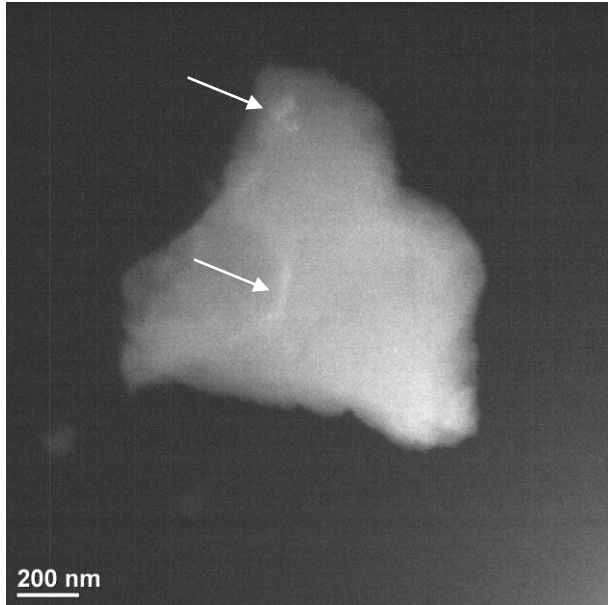


- This TEM brightfield image shows a particle of sample P03 at a magnification of 10.000 times.
- The diffraction pattern does not show any pronounced crystalline reflexes.
- However ...



TEM investigation

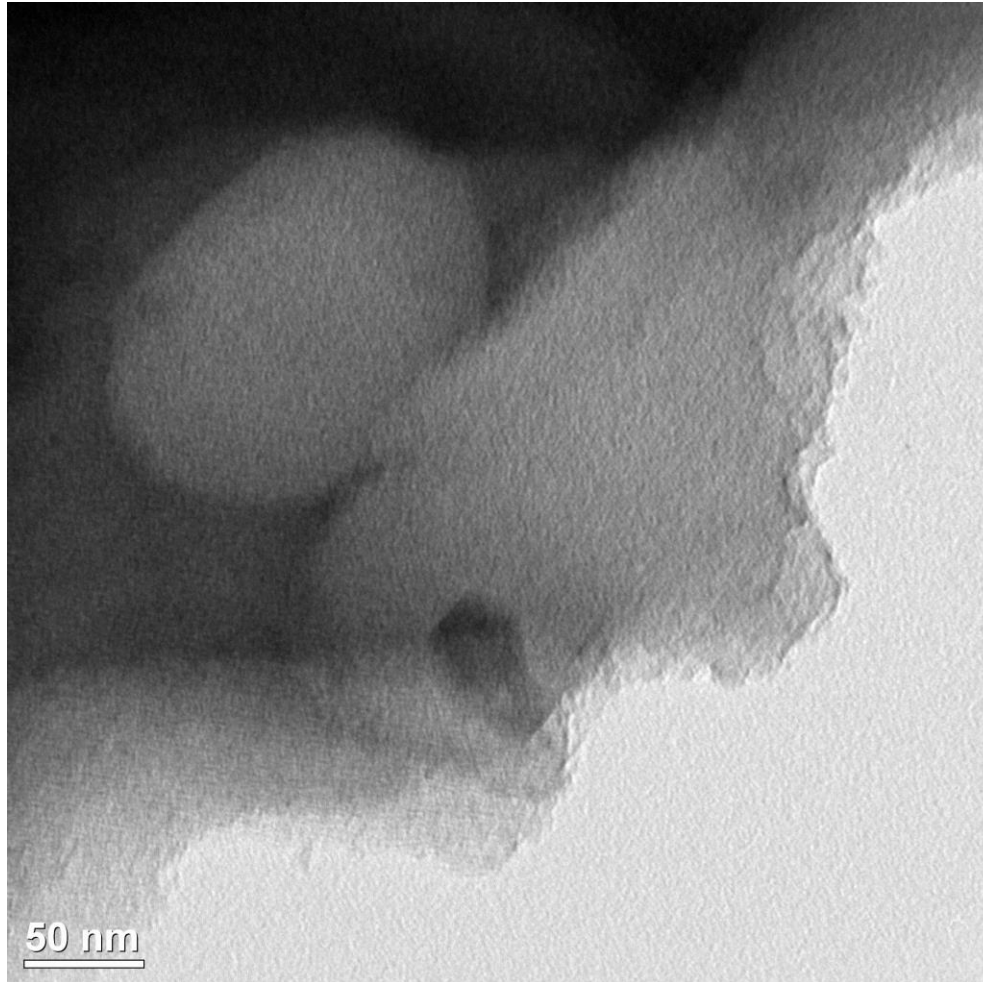
Sample P03 (MB2976A)



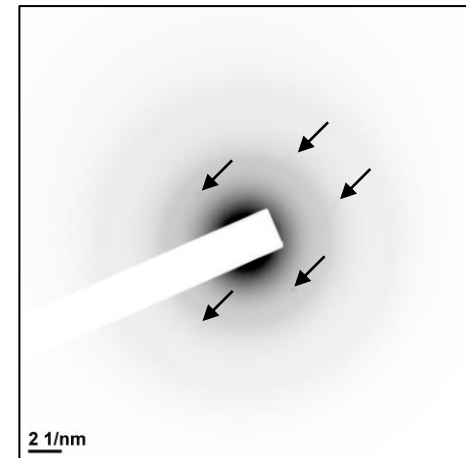
- ... darkfield imaging at 10.000 times magnification reveals hints of crystallinity (arrows).
- This effect can also be observed with other particles.

TEM investigation

Sample P03 (MB2976A)

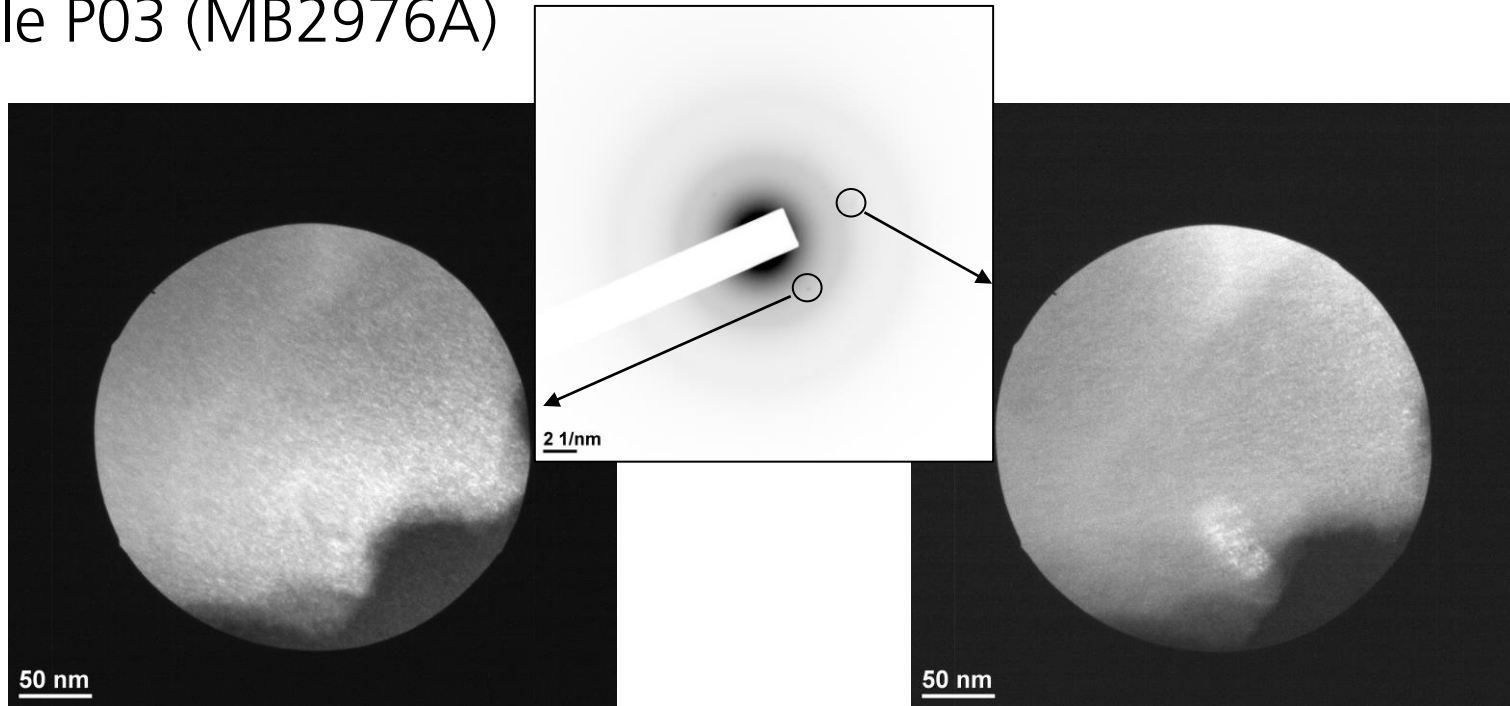


- This TEM brightfield image shows a region of a particle of sample P03 at a magnification of 50.000 times that exhibited no pronounced crystalline reflexes when the diffraction pattern was taken from the whole particle.
- However, the imaged region indeed shows weak reflexes (arrows).



TEM investigation

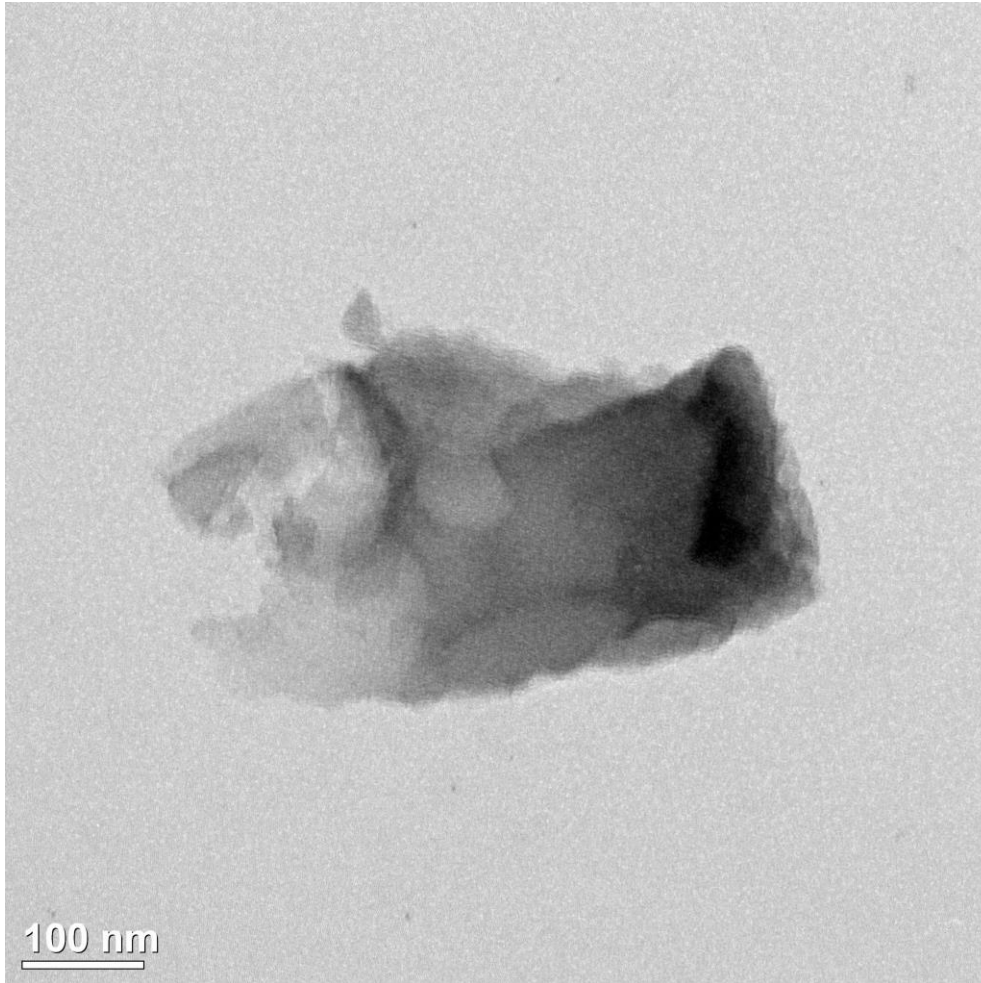
Sample P03 (MB2976A)



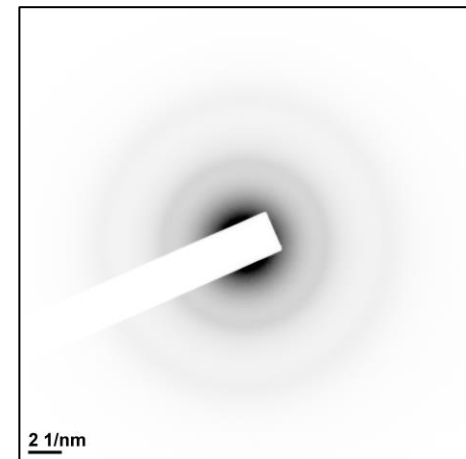
- Darkfield images (50.000 times magnification) show the crystalline areas belonging to the indicated reflexes.
- It seems that the particle has a nano-crystalline structure (at least in this area).

TEM investigation

Sample P03 (MB2976A)



- This TEM brightfield image shows a particle of sample P03 at a magnification of 10.000 times.
- A porous structure is visible.
- The diffraction pattern (of the left part of the particle) does not show any pronounced crystalline reflexes.



Final Remarks

- The analyses are carried out by specifically qualified employees and by the use of up-to-date analysis machines. Nevertheless, we are not able to provide guarantee for the correctness of the drawn conclusion.
- The results only refer to the provided and examined samples.
- Sample taking and shipping of the samples to the ZAA took place under the responsibility of the client.
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